Multimodal Deep Learning

Exercise Sheet 6

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Exercise 1: Information-Theory Basics

Symbols w are drawn from a finite alphabet W. Let $p(w)$ be the *true* distribution and $q_\theta(w)$ a model family parametrized by θ.

(a) Which information-theoretic quantity should be *minimized* to make q_θ approach the true distribution p?
Hint: Two equivalent answers are valid under the support assumption.

(b) Explain why the quantity from (a) is a sensible objective and show the explicit connection

$$H(p, q_\theta) = -\mathbb{E}_p[\log q_\theta(w)].$$

Why does this imply that *maximizing* the (average) log-likelihood is equivalent to *minimizing* cross-entropy?

(c) Show that

$$H(p, q_\theta) = H(p) + D_{\mathrm{KL}}(p \| q_\theta).$$

(d) Analyse what goes wrong in the argument of (c) if $q_\theta(w) = 0$ for some w with $p(w) > 0$.
Hint: Discuss both the mathematical value and its practical implications for optimization.

Empirical setting Draw n i.i.d. samples $w_{1:n} \sim p$ and let $\hat{p}_n$ be their empirical distribution. Show that

$$\mathrm{NLL}(\theta) = -\frac{1}{n}\sum_{i=1}^{n} \log q_\theta(w_i) \;=\; H(\hat{p}_n, q_\theta),$$

and hence that minimizing NLL is the same as minimizing $D_{\mathrm{KL}}(\hat{p}_n \| q_\theta)$.
Remark: Explain why driving this empirical KL to zero need not guarantee good *generalization* to unseen data.

(e) **Mutual information forms**
For $(X, Y) \sim p_{XY}$, starting from $I(X;Y) = D_{\mathrm{KL}}(p_{XY} \| p_X p_Y)$:

1. Derive the equivalent entropy form $I(X;Y) = H(X) + H(Y) - H(X, Y)$.

2. Give a one-sentence interpretation of the case $I(X;Y) = 0$.

(f) Numerical sanity check Let $W = \{0, 1\}$ with $p(1) = 0.9$, $p(0) = 0.1$ and model $q_\theta(1) = 0.8$, $q_\theta(0) = 0.2$. Using base-2 logarithms compute:

1. $H(p)$,
2. $H(p, q_\theta)$,
3. $D_{\mathrm{KL}}(p \| q_\theta)$,

and verify numerically that $H(p, q_\theta) = H(p) + D_{\mathrm{KL}}(p \| q_\theta)$.

Exercise 2: Contrastive Learning

Throughout, fix a **mini-batch size** N and two encoders

$$f_\theta : \mathcal{X} \to \mathbb{R}^d, \qquad g_\phi : \mathcal{X} \to \mathbb{R}^d.$$

For each data pair (x_i, x_i') we write the embeddings as

$$z_i = f_\theta(x_i), \qquad z_i' = g_\phi(x_i'), \qquad i = 1, \dots, N.$$

For any $a, b \in \mathbb{R}^d$ define

$$s(a, b) = \frac{\langle a, b\rangle}{\|a\|_2 \, \|b\|_2}, \qquad \tau > 0 \text{ (temperature).}$$

We collect the **logits** in the matrix

$$L \;=\; [\ell_{ij}]_{i,j=1}^N, \qquad \ell_{ij} = \frac{s(z_i, z_j')}{\tau}.$$

(a) **Soft-max connection**

(a) **Row-wise soft-max.** For a fixed *anchor row* i turn the logits $(\ell_{i1}, \dots, \ell_{iN})$ into the probability distribution

$$\boxed{p_\theta(j \mid i) = \mathrm{softmax}_j(\ell_{ij}) = \frac{e^{\ell_{ij}}}{\sum_{k=1}^N e^{\ell_{ik}}}}.$$

Show that the original InfoNCE objective

$$\mathcal{L}_{\text{InfoNCE}} = -\frac{1}{N} \sum_{i=1}^N \log p_\theta(j = i \mid i)$$

equals the **cross-entropy**

$$\boxed{\mathcal{L}_{\text{InfoNCE}} = \frac{1}{N} \sum_{i=1}^N \mathrm{CE}\big(q(\cdot \mid i),\, p_\theta(\cdot \mid i)\big)}, \qquad q(j \mid i) = \mathbf{1}\{j = i\}.$$

(b) **Negatives.** Why are the non-matching pairs (z_i, z_j') with $j \neq i$ called *negatives*? Hint: look at the zero entries of $q(\cdot \mid i)$.

(b) **Symmetric loss (CLIP)**

CLIP averages two InfoNCE terms:

$$\mathcal{L}_{\text{CLIP}} = \tfrac{1}{2}\big[\mathcal{L}_{\text{I}\to\text{T}} + \mathcal{L}_{\text{T}\to\text{I}}\big].$$

(a) **Image → Text.** Show

$$\mathcal{L}_{\text{I}\to\text{T}} = \frac{1}{N} \sum_{i=1}^N \mathrm{CE}\big(q(\cdot \mid i),\, p_\theta(\cdot \mid i)\big),$$

i.e. the cross-entropy over the *rows* of L (identical to Part A).

(b) **Text → Image.** Treat each *column* j as the anchor, define

$$p_\theta(i \mid j) = \frac{\exp(\ell_{ij})}{\sum_{k=1}^{N} \exp(\ell_{kj})}, \qquad q(i \mid j) = \mathbf{1}\{i = j\},$$

and show the analogous formula for $\mathcal{L}_{\mathrm{T}\to\mathrm{I}}$.

(c) **All $2N-2$ negatives.** Conclude that $\mathcal{L}_{\mathrm{CLIP}}$ can be written as a *single* loss where both rows and columns of L are soft-max normalised, so every positive pair (i, i) is contrasted against all other entries in its row and column.

(c) **Uniformity versus Alignment**

Following Wang & Isola (2020) define

$$\mathcal{A} = \mathbb{E}\big[\|Z - Z'\|_2^2\big], \qquad \mathcal{U} = \log \mathbb{E}\big[\exp(-2\|Z - \tilde{Z}\|_2^2)\big],$$

where $\tilde{Z}$ is an independent sample from p_Z.

(a) Argue qualitatively how minimising $\mathcal{L}_{\mathrm{InfoNCE}}$ *decreases* $\mathcal{A}$ (brings paired points together) and *decreases* $\mathcal{U}$ (spreads all representations).

(b) Explain why this trade-off becomes harder when the batch contains many modalities or when N is very large.

Exercise 3: Variational Auto-Encoders (VAE) and VQ-VAE

The VAE loss is $\mathcal{L}_{\mathrm{VAE}} = \mathcal{L}_{\mathrm{rec}} + \beta\, D_{\mathrm{KL}}[q(z|x)\|p(z)]$.

(a) **Role of β**

Describe the trade-off controlled by β. What happens if $\beta = 0$? What happens when β is very large?

(b) **Closed-form KL (1-D case)**

Assume a one-dimensional standard normal prior $p(z) = \mathcal{N}(0, 1)$ and an approximate posterior $q_\phi(z \mid x) = \mathcal{N}(\mu(x), \sigma^2(x))$. Show that

$$D_{\mathrm{KL}}(q_\phi(z \mid x) \,\|\, p(z)) \;=\; \tfrac{1}{2}\big(\mu^2 + \sigma^2 - \log \sigma^2 - 1\big).$$

Hint: write out the KL, collect quadratic and logarithmic terms, and simplify.

(c) **Analogy to VQ-VAE**

In a VQ-VAE the continuous latent vector $z_e(x) \in \mathbb{R}^d$ produced by the encoder is *quantized* to the nearest codebook vector $e_k \in \mathbb{R}^d$:

$$z_q(x) \;=\; e_{k^\star}, \quad k^\star = \arg\min_k \big\|z_e(x) - e_k\big\|_2.$$

The loss has three terms

$$\mathcal{L}_{\text{VQ-VAE}} \;=\; \mathcal{L}_{\mathrm{rec}} + \underbrace{\|\mathrm{sg}[z_e(x)] - e_{k^\star}\|_2^2}_{\text{codebook}} + \beta\, \underbrace{\|z_e(x) - \mathrm{sg}[e_{k^\star}]\|_2^2}_{\text{commitment}},$$

where $\mathrm{sg}[\cdot]$ denotes the "stop-gradient" operator.

(i) Explain how the *codebook* term plays a role analogous to the KL divergence in a continuous VAE.

(ii) Explain the purpose of the *commitment* term and the effect of the hyper-parameter β in this context.

Exercise 4: Generative Self-Supervision Paradigms

(a) **Autoregressive modelling**
Derive the cross-entropy objective Show that maximising the autoregressive likelihood

$$\max_\theta \prod_{t=1}^{N} p_\theta(x_t \mid x_{<t})$$

of a discrete token sequence $x_{1:N}$ is equivalent to minimising the *average token-level cross-entropy loss*

$$\mathcal{L}(\theta) \;=\; -\frac{1}{N}\sum_{t=1}^{N} \log p_\theta(x_t \mid x_{<t}).$$

Interpret perplexity On a held-out set, *perplexity* can be written as

$$\mathrm{PPL} \;=\; \exp\bigl(\mathcal{L}(\theta)\bigr).$$

Interpret the meaning of a perplexity of

(i) 1,
(ii) equal to the vocabulary size $|V|$, and
(iii) halfway between those extremes.

(b) **Masked image modeling (MIM)**
For a Vision Transformer with patch size 16×16, explain how patch masking affects the spatial receptive field. Design a simple experiment to measure reconstruction quality as a function of the mask ratio. Why do contemporary works often choose $\rho \approx 0.75$?

Exercise 5: Multimodal Alignment - Understanding

(a) **Positives and negatives in CLIP**
For a batch of N image-text pairs (I_i, T_i), how many positive and how many negative pairs are considered for a *single* image I_i when computing the image-to-text loss component?

(b) **Extending CLIP to three modalities**
Suppose you add an audio encoder producing embeddings A_i. Formulate a contrastive loss that aligns all three modalities (I, T, A). How many distinct pairwise loss terms are required in total? Compare your idea to OmniBind from the lecture.

(c) **Sigmoid loss (SigLIP)**
Unlike CLIP, SigLIP replaces the softmax with a pairwise sigmoid loss. Write the SigLIP loss for one positive pair (I_i, T_i) and explain why negatives are sampled differently compared to CLIP. What happens to the influence of easy vs hard negatives on the image and text encoders' updates?

(d) **JEPA versus CLIP**
Briefly compare the objective of a Joint Embedding Predictive Architecture (JEPA) with CLIP. Why is JEPA considered predictive rather than contrastive? List two key advantages JEPA claims over contrastive approaches, and briefly explain how its architecture enables each advantage.

(e) **JEPA Masking** Compare the effects of using a single large contiguous mask versus many small scattered masks on the learned representation. How might the optimal mask strategy differ between images and audio waveforms?

(f) **MultiMAE**
MultiMAE masks a random subset of modalities and reconstructs the missing ones. Formulate the training objective for a three-modal setup (I, T, A) and discuss how it differs from both CLIP and JEPA.

(g) **MultiMAE Loss** MultiMAE linearly combines per-modality reconstruction losses. What could happen if one modality's loss dominates the total objective?

(h) **MultiMAE Shared encoder vs. modality-specific decoders**

Give two reasons why the authors attach small, separate decoders instead of one large shared decoder. What possible downside might arise when decoders are too lightweight or too isolated from each other?